

FAT LOSS SCIENCE

CARDIO PROTOCOLS

FAT BURNER STACK

CONTEST PREP

ULTIMATE FAT LOSS MASTERCLASS

Energy Balance · Macros · Cardio Science · Fat Burners · Metabolic Adaptation · Contest Prep

Evidence-based fat loss from physique athletes and sports science research

ELITE FITNESS GUIDE SERIES

Research-Based · Professional Edition · 2024

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■ 1. THE SCIENCE OF FAT LOSS

Fat loss is governed by one unbreakable law: **energy balance**. You must consume less energy than you expend. However, the *quality* of your deficit — which macros, what types of training, and how you manage metabolic adaptation — determines whether you lose fat vs muscle.

■ A meta-analysis of 53 weight loss trials (Sacks 2009, NEJM) found that the specific macronutrient composition of caloric-deficit diets mattered far less than overall energy deficit for weight loss. However, high-protein diets (>1.2g/lb) consistently preserved significantly more lean mass during deficit.

The Three Laws of Fat Loss:

- **Law 1 — Energy Deficit:** 1 lb of fat = approximately 3,500 kcal. A 500 kcal/day deficit = ~1 lb fat loss/week theoretically. In practice, metabolic adaptation reduces this. Aim for 0.5–1 lb/week for maximum muscle preservation
- **Law 2 — Protein Priority:** High protein preserves lean mass in deficit. Protein has the highest thermic effect (25–30% of calories burned in digestion). Minimum 1.0 g/lb body weight; optimal 1.5–2.0 g/lb during aggressive cuts
- **Law 3 — Training Stimulus:** Resistance training while in caloric deficit sends a 'survival' signal to preserve muscle. Without resistance training, 25–40% of weight loss comes from muscle mass

■ 2. CALCULATING YOUR DEFICIT — TDEE & BMR

Step 1 — Calculate BMR (Mifflin-St Jeor equation):

Men: $BMR = 10 \times \text{weight(kg)} + 6.25 \times \text{height(cm)} - 5 \times \text{age} + 5$

Women: $BMR = 10 \times \text{weight(kg)} + 6.25 \times \text{height(cm)} - 5 \times \text{age} - 161$

Activity Level	Multiplier	Description
Sedentary	$BMR \times 1.2$	Desk job; no exercise
Light Activity	$BMR \times 1.375$	1–3 workouts/week
Moderate Activity	$BMR \times 1.55$	3–5 workouts/week
Very Active	$BMR \times 1.725$	Hard training 6–7 days/week
Extremely Active	$BMR \times 1.9$	Athlete or physical labor + daily training

Deficit Recommendations by Goal:

Goal	Deficit	Weekly Fat Loss	Muscle Preservation
Slow Recomp	100–200 kcal	0.1–0.2 lbs	Excellent (muscle gain possible)
Moderate Cut	300–400 kcal	0.5–0.7 lbs	Very good (minimal loss)
Aggressive Cut	500–600 kcal	0.8–1.0 lbs	Good (high protein critical)
Show Cut	600–800 kcal	1.0–1.5 lbs	Moderate (AAS/SARMs + high protein essential)

■ 3. MACRONUTRIENT STRATEGY FOR FAT LOSS

Macro	Fat Loss Target	Why	Food Sources
Protein	1.5–2.0 g/lb BW	Muscle preservation + satiety + TEF (25–30% burned in digestion)	Chicken, fish, eggs, Greek yogurt, cottage cheese, tofu, paneer
Fats	0.35–0.5 g/lb BW	Hormonal health; fat-soluble vitamins; satiety; never go below 40g/day total	Olive oil, avocado, nuts, eggs, fatty fish
Carbohydrates	Fill remaining calories	Primary energy for training performance; not the enemy — only context matters	Oats, sweet potato, rice, fruits, vegetables
Fibre	35–50 g/day	Gut health; slows digestion; significantly improves satiety; improves insulin sensitivity	All vegetables, legumes, oats, psyllium husk

Carb Cycling — Advanced Strategy: On training days eat higher carbs (around workouts); on rest days reduce carbs. This optimizes muscle glycogen while increasing fat burning on low-activity days. Example: 200g carbs on training days; 80–100g on rest days.

■ 4. CARDIO SCIENCE — LISS vs HIIT vs ZONE 2

All three cardio modalities burn calories and accelerate fat loss — the differences are in sustainability, muscle preservation, and metabolic stress:

Type	Heart Rate Zone	Duration	Calories (approx)	Best Use Case	Muscle Preservation
LISS (Low Intensity Steady State)	50–65% max HR	30–60 min	200–350 kcal	Daily fat-burning maintenance; fasted cardio; recovery days	Excellent — minimal muscle stress
Zone 2 (Aerobic Base)	65–75% max HR	45–90 min	300–500 kcal	Mitochondrial development; metabolic flexibility; endurance base	Very good
HIIT (High Intensity Interval)	80–95% max HR	15–25 min	200–400 kcal (+ EPOC)	2–3×/week max; high calorie afterburn (EPOC); time-efficient	Poor if excessive — avoid >3×/week in deficit
Sprint Intervals	95–100% max HR	10–15 min	250–400 kcal (+ EPOC)	Weekly power maintainer; high EPOC; preserves fast-twitch muscle	Good — if brief and infrequent

■ Greer 2015 meta-analysis: HIIT and LISS produce equivalent fat loss when total calories burned are equated. HIIT is time-efficient but higher injury risk and greater muscle catabolism risk in caloric deficit. Recommendation: LISS primary (4–5×/week) + HIIT secondary (1–2×/week).

■ 5–6. METABOLIC ADAPTATION & REFEEDS

Metabolic Adaptation is the reduction in metabolic rate that occurs in response to caloric deficit. The body adapts to reduce energy expenditure, protecting against starvation. This is why fat loss plateaus — what was once a 500 kcal deficit becomes 200 kcal or less over weeks.

Four Components of Metabolic Adaptation:

- **Reduced BMR** (~5–10%): Fewer calories needed for basic body functions as body weight drops
- **Adaptive thermogenesis** (~5–15%): Beyond what weight loss predicts — body actively reduces heat production
- **Reduced NEAT** (~10–25%): Non-exercise activity thermogenesis drops (you fidget less, move less instinctively)
- **Reduced exercise efficiency** (~5%): Body becomes more efficient at exercise, burning fewer calories for same work

Refeed Strategy (Targeted Leptin Restoration):

- Leptin is the primary satiety hormone and metabolic regulator — it drops 50% within 7 days of caloric deficit
- A high-carbohydrate refeed day (TDEE + 20–30% surplus from carbs ONLY) temporarily restores leptin, reversing some adaptive thermogenesis
- Frequency: 1 refeed day per week during aggressive cuts; 1 per 2 weeks during moderate cuts
- Refeed protocol: Increase carbohydrates to 3–4× normal intake; keep protein HIGH; keep fats LOW (fat doesn't restore leptin)
- Diet break (1–2 weeks at TDEE): More powerful than refeeds for reversing adaptation; every 8–12 weeks on extended cuts

■ 7. NATURAL FAT BURNER STACK — EVIDENCE-BASED

Supplement	Dose	Mechanism	Evidence Level
Caffeine (anhydrous)	3–6 mg/kg body weight	Increases adrenaline (epinephrine) release → mobilizes fat stores (lipolysis); increases BMR 3–11%; synergistic with all other fat burners	Very strong — most researched fat loss compound
Green Tea Extract (EGCG)	400–500 mg EGCG/day	Inhibits COMT enzyme — prolongs noradrenaline activity; additive fat-burning effect with caffeine	Strong — multiple meta-analyses confirm 3–5% additional fat loss vs placebo
L-Carnitine	2–3 g/day (L-Carnitine L-Tartrate)	Transports fatty acids into mitochondria for oxidation — rate-limiting step in fat metabolism. Works best with carbohydrates	Moderate — requires adequate tissue carnitine saturation over 4+ weeks
Yohimbine HCl	0.2 mg/kg body weight, fasted	Alpha-2 adrenergic receptor antagonist — blocks the 'anti-fat-burning' receptors found in stubborn fat areas (lower abdomen, hips, thighs)	Moderate — significant for fasted cardio; no effect with elevated insulin
Synephrine (Bitter Orange)	10–20 mg	Adrenergic agonist (beta-3 receptor) — increases metabolic rate without the cardiovascular side effects of ephedrine	Moderate — gentler alternative to ephedrine; synergistic with caffeine
CLA (Conjugated Linoleic Acid)	3.2 g/day	Reduces fat cell (adipocyte) differentiation; mild fat oxidation increase	Weak-moderate — modest effects (0.5 lb/month vs placebo)

Optimal Natural Stack Timing: Caffeine + EGCG fasted 30 min before cardio. Yohimbine fasted with cardio (insulin blocks Yohimbine's effect). L-Carnitine + carbohydrates. Take caffeine by 2 PM to avoid sleep disruption.

■ 8–9. CLENBUTEROL, ECA & T3

ECA Stack (Ephedrine + Caffeine + Aspirin):

- **Ephedrine:** 20–25 mg + **Caffeine:** 200 mg + **Aspirin:** 81 mg — taken 2–3× daily
- Synergistic mechanism: Ephedrine (beta-adrenergic agonist) increases noradrenaline; caffeine blocks adenosine AND inhibits phosphodiesterase (prolongs cAMP effect); aspirin (prostaglandin inhibition) extends the thermogenic response
- Cycle: 3 weeks ON (build tolerance) / 1 week OFF. Never exceed 3× daily dose
- Avoid if: hypertension, heart conditions, anxiety disorders, thyroid disease

Clenbuterol (see PDF 01 for full protocol): 20→120 mcg/day, 2 weeks on/off. Beta-2 agonist — thermogenic + mild anabolic. More powerful than ECA; more side effects.

T3 Thyroid Protocol (see PDF 01 for full details): 25→50→25 mcg/day tapered protocol. Dramatically increases metabolic rate. Mandatory taper — cold turkey causes hypothyroid crash.

■ 10–12. CONTEST PREP · PEAK WEEK · MAINTENANCE

12-Week Contest Prep Framework:

Weeks	Deficit	Cardio Volume	Compounds (optional)	Focus
12–10	500 kcal	30 min LISS 4×/wk	None / natural stack	Establish discipline; set macro targets
9–7	500–600 kcal	40 min LISS 5×/wk	Clen/ECA if desired	Accelerate fat loss; refeed 1×/week
6–4	500–700 kcal	45 min LISS 6×/wk + 2× HIIT	T3 25–50 mcg if using	Address stubborn fat; diet break optional
3–2	600–700 kcal	50 min 2×/day or 60 min 1×/day	Peak compounds if using	Final detail; reduce water-retaining foods
1 (Peak Week)	Carb deplete → Carb load	Minimal (just pump sessions)	Diuretics only if very experienced	See Peak Week protocol below

Peak Week Protocol:

- Days 7–5 (Depletion): Reduce carbs to 50g/day; increase cardio to 2×/day; glycogen depletes for max carb loading response
- Days 4–3 (Loading): 600–800g carbs/day from clean sources; reduce training volume dramatically
- Day 2: Moderate carbs (400g); assess fullness; adjust if needed
- Day 1 (Show Day): Small carb and sodium doses with good food; stay hydrated; avoid new foods
- Water: Drink normally until night before; slight reduction day of show. Never extreme dehydration — dangerous and rarely improves conditioning

Maintaining Results (Post-Cut Reverse Diet): Increase calories by 50–100 kcal/week over 8–12 weeks back to maintenance TDEE. Rapid post-cut calorie increase triggers fat regain ('rebound'). Reverse dieting allows metabolic rate to normalize while minimizing fat regain.